



Construction ; 8.0" x 8.0" single sided PCB with gold plated 0.1" edge connector pads.

3. CIRCUIT DESCRIPTION

The EPROMs are arranged as two blocks, X1-4 and X10-13, each block may be set to take either 1k x 8 2708's or 2k x 8 2716's (+5V only versions e.g. Intel 2716 or Texas 2516).

Note that 2708's and 2716's cannot be mixed within one block. Thus X1-4 must be either all 2708's or all 2716's, and X10-13 must also be either all 2708's or 2716's. In the following description, where signals used for 2708's differ from those used for 2716's, then the 2716 alternative is shown in brackets.

All address inputs to the board A0-A15 are buffered by the permanently enabled X15,16, then the least significant 10(11) lines A0-9(10) are connected to all EPROM.

The high order address bits A12(13)-15 are fed to X5,7 where they are compared to the switch settings of S1a-S1d(c) and S2a-S2d(c) to determine if the address present on the system bus is within the 4(8)k byte range selected for either EPROM block. If this is so then X5 or X7 o/p pin 3,6 etc. go high, enabling one of the X6 gates.

If this happens while the board R/W input is high (signifying a 'Read' operation) then X6 pin 6, or pin 8, will go low and enable one of the two one-out-of-four decoders in X9.

The A & B inputs of X9 are connected to the remaining two address lines A10,11 (A11,12); thus a 'Read' operation from an address recognised by the board will cause one of the 1/4 decoder outputs to go low. These outputs are connected via switches S1e-h and S2e-h to the 'select' inputs of EPROMS X1-4 and X10-13, and are also connected to inputs of the NOR gate X8, causing its output to go high, enabling X6. Then, when the system bus 'E' signal goes high, the data output buffer X14 is enabled to transfer data from the selected EPROM onto the system bus.

If an EPROM is not fitted in any one of the X1-4, X10-13 positions, then the corresponding switch S1e-h or S2e-h should be open-circuit. This will prevent the data output buffer X14 from being enabled when the 'vacant' 1(2)k byte address space is accessed, allowing this address space to be used for memory or I/O elsewhere in the system.

An on-board regulator is fitted to derive -5V from the system -12V rail when 2708's are fitted.

NOTE; Some types of 2708 are inclined to lose data if left with positive supplies connected but without the -5V supply. The most likely cause of a loss of the -5V supply is loss of the system -12V rail, thus it is best to arrange the system power supply so that the positive supplies are switched off (possibly via a relay) on failure of the -12V rail.

4. CONSTRUCTION

Refer to Figs 2,3,4

The close spacing of the tracks on this board means that great care has to be taken to avoid accidental shorts between tracks due to, for example, solder splashes. The constructor should therefore take care to make good soldered joints using the minimum amount of solder whilst ensuring that a joint is made all round the component pin or wire end. Use of a small iron is essential. Also, before fitting any component, feel all over the track side of the board for any loose copper swarf left from the board drilling process, and examine it carefully for unetched copper 'bridges' between tracks.

IC sockets should be fitted in the X1-4, X10-13 positions to cater for EPROM changes, whether sockets are used for the other IC's or not is a matter of personal preference.

The best order of construction is;

First decide whether 2708's or 2716's are to be fitted in each block (X1-4 and X10-13). Then fit all straps (see Figs 2&4) using sleeved wire where appropriate.

Next, fit the resistors, followed by the IC's (except for the EPROM's X1-4 & X10-13 which should not be fitted at this stage), and the EPROM sockets.

Finally, mount the DIL switches, capacitors and - if required - the -5V regulator with a suitable small heatsink.

After assembly clean the track side of the board thoroughly with a small stiff brush (a very hard toothbrush is ideal) and examine it carefully for short circuits caused by bent leads, excess solder on joints, solder splashes etc.

5. TESTING

Before plugging in any EPROMs make a thorough visual check to ensure that the correct components have been fitted, that the ICs and capacitors are the right way round (some IC's have a 'notch' at both ends, in which case look for a small hole indicating pin 1), and that the correct straps have been fitted.

Then check carefully with an ohm-meter for short circuits between tracks, especially those connecting X1-4 & X10-16, also between 0 & +5V lines. (Note that some reading is to be expected due to the internal resistances of the ICs). This stage is most important as it can reveal faults which would otherwise be extremely difficult to locate.

Apply +5 (and +, -12V if used) to the board and leave it for a few minutes while checking for signs of overheating. Check the derived -5V line.

Set switches S1e-h, S2e-h to the open position (i.e. no EPROM fitted) and set the address selection switches S1a-d, S2a-d to an address range which is already used by some other memory in the system.

Plug the board into the system and check to see that it doesn't affect normal operation in any way, particularly accessing the other memory in the same address space.

Remove power, then insert an EPROM with known contents and reset the address and EPROM select switches as appropriate. Reconnect the power while monitoring the -5V rail (if used). It is worth while checking the -5V rail every time power is applied after an EPROM change, as loss of the -5V can affect the contents of some types of 2708.

Check that the correct data can be read from the EPROM.

77-68 ROM-A

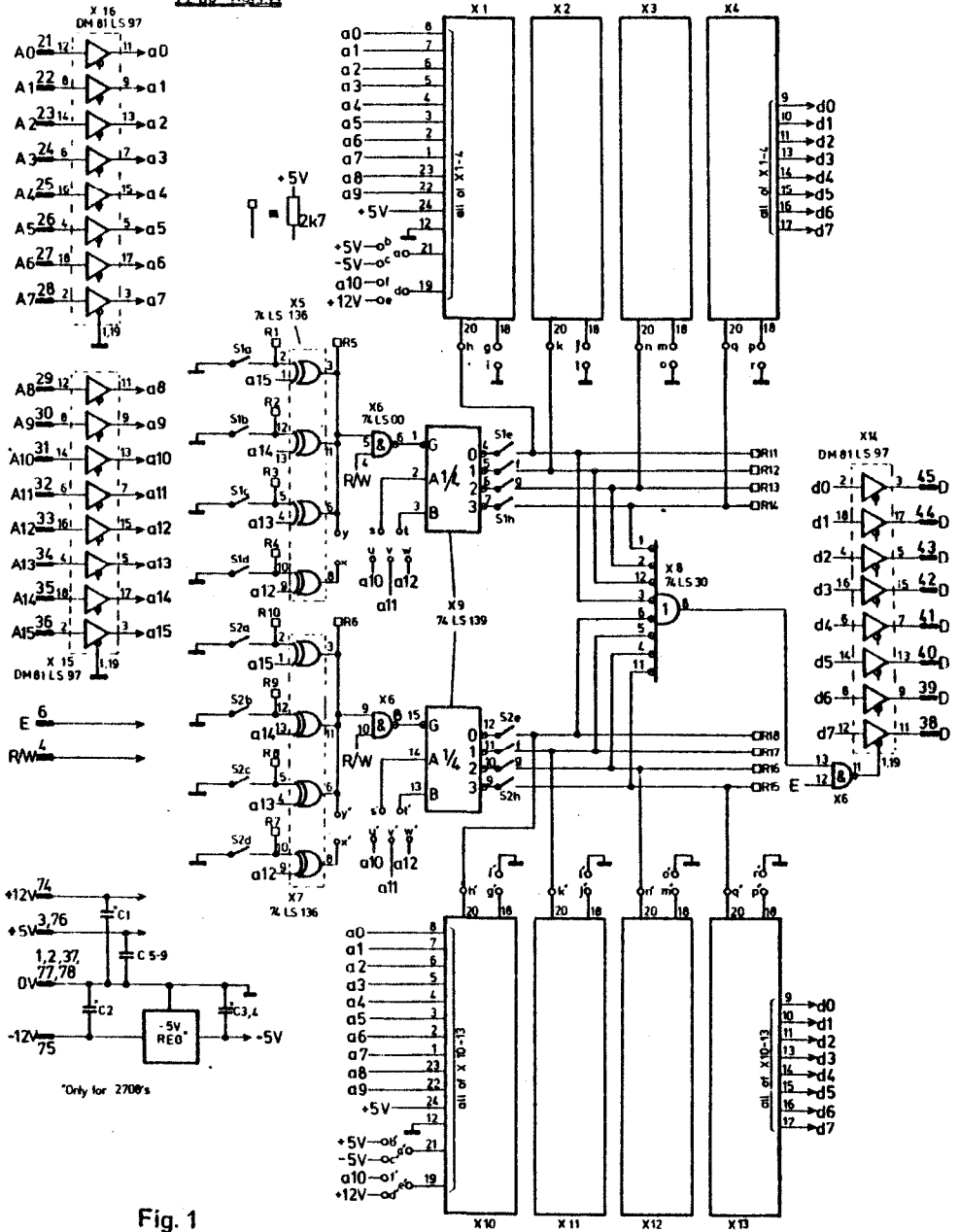
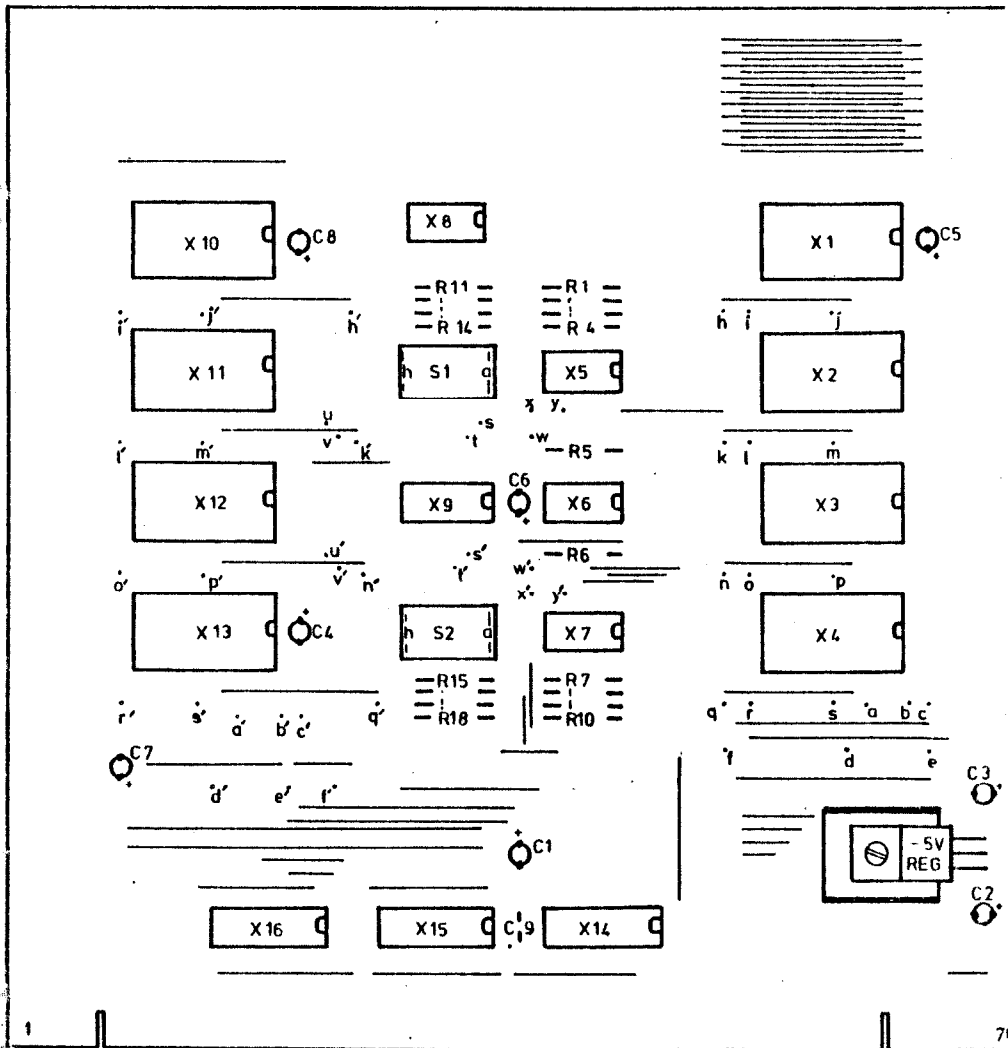


Fig. 1



77-68 ROM.A BOARD
Component Side View

For connection to lettered points (a,b,a'etc.) see Fig. 4

Fig 2

77-68 ROM-A BOARD

Components

IC's

X1,2,3,4,10,11,12,13	2708 or Intel (5V) version of 2716 EPROM
X5,7	74LS136 quad exclusive or, o/c o/p
X6	74LS00 quad 2 i/p nand
X8	74LS30 eight i/p nand
X9	74LS139 dual 1/4 decoder
X14,15,16	DM81LS97 octal tri-state buffer
-5V reg	LM320T-5 or LM320MP-5

Resistors

R1-18 2k7 ohm miniature

Capacitors

C1,8 4 μ 7 16V tantalum bead
C9 0 μ 1 ceramic

Misc

Sl,2;8 pole single throw on-off DIL switches (2 off)
DIL sockets; at least 8 off 24 pin for X1-4,10-13
 plus, if desired; 3 off 20 pin
 1 off 16 pin
 4 off 14 pin

Small heat-sink for -5V regulator

77-68 ROM-A printed circuit board

Mating edge connector socket; 0.1" single sided 78 way (76 connections plus polarising keys)

Note

C1-4 and -5V regulator are only required if 2708's fitted.

Fig 3

Optional Strapping

EPROM block A; X1-4

For 2708's strap; a-c, d-e, g-i, j-l, m-o, p-r, s-u, t-v, x-y
leave b, f, h, k, n, q, w, open

For 2716's strap; a-b, d-f, g-h, j-k, m-n, p-q, s-v, t-w
leave c, e, i, l, o, r, u, x, y, open

EPROM block B: X10-13

As for block A but a', b', etc instead of a, b etc.

Address Switch Setting

S1 is used for block A (X1-4), S2 for block B (X10-13)

S1(2)				Address range for 2708's			
a	b	c	d	X1(X10)	X2(X11)	X3(X12)	X4(X13)
0	0	0	0	0000-03FF	0400-07FF	0800-0BFF	0C00-0FFF
0	0	0	1	1000-13FF	1400-17FF	1800-1BFF	1C00-1FFF
0	0	1	0	2000-23FF	2400-27FF	2800-2BFF	2C00-2FFF
0	0	1	1	3000-33FF	3400-37FF	3800-3BFF	3C00-3FFF
0	1	0	0	4000-43FF	4400-47FF	4800-4BFF	4C00-4FFF
0	1	0	1	5000-53FF	5400-57FF	5800-5BFF	5C00-5FFF
0	1	1	0	6000-63FF	6400-67FF	6800-6BFF	6C00-6FFF
0	1	1	1	7000-73FF	7400-77FF	7800-7BFF	7C00-7FFF
1	0	0	0	8000-83FF	8400-87FF	8800-8BFF	8C00-8FFF
1	0	0	1	9000-93FF	9400-97FF	9800-9BFF	9C00-9FFF
1	0	1	0	A000-A3FF	A400-A7FF	A800-ABFF	AC00-AFFF
1	0	1	1	B000-B3FF	B400-B7FF	B800-BBFF	BC00-BFFF
1	1	0	0	C000-C3FF	C400-C7FF	C800-CBFF	CC00-CFFF
1	1	0	1	D000-D3FF	D400-D7FF	D800-DBFF	DC00-DFFF
1	1	1	0	E000-E3FF	E400-E7FF	E800-EBFF	EC00-EFFF
1	1	1	1	F000-F3FF	F400-F7FF	F800-FBFF	FC00-FFFF

S1(2)				Address range for 2716's			
a	b	c	d	X1(X10)	X2(X11)	X3(X12)	X4(X13)
0	0	0	X	0000-07FF	0800-0FFF	1000-17FF	1800-1FFF
0	0	1	X	2000-27FF	2800-2FFF	3000-37FF	3800-3FFF
0	1	0	X	4000-47FF	4800-4FFF	5000-57FF	5800-5FFF
0	1	1	X	6000-67FF	6800-6FFF	7000-77FF	7800-7FFF
1	0	0	X	8000-87FF	8800-8FFF	9000-97FF	9800-9FFF
1	0	1	X	A000-A7FF	A800-AFFF	B000-B7FF	B800-BFFF
1	1	0	X	C000-C7FF	C800-CFFF	D000-D7FF	D800-DFFF
1	1	1	X	E000-E7FF	E800-EFFF	F000-F7FF	F800-FFFF

Note; S1(2)a-d; '0' = open circuit, '1' = closed, 'X' = don't care

S1e-h & S2e-h settings;

EPROM fitted;	X1	X2	X3	X4	X10	X11	X12	X13
Switch closed;	S1e	S1f	S1g	S1h	S2e	S2f	S2g	S2h
other switches to be open circuit								

Fig. 4

77-68 ROM.A Appendix

Use of other ROMs

Note; the following has been derived from data sheets, not from experiment, and is therefore subject to proof by trial.

INTEL	1702	Not useable	
"	2704	512 x 8 EPROM	Strap as 2708. Accessible in lower half of 1k address space only (A9=0).
"	2308	1024 x 8 ROM	Strap as 2708 but pin 18 (CS2) must be taken to 0V or +5V depending on chip programming.
"	2316A	Not useable	
"	2316E	2048 x 8 ROM	Use as 2716
"	2608	1024 x 8 PROM	Use as 2708
"	2616	2048 x 8 PROM	Use as 2716
"	8708	1024 x 8 EPROM	Use as 2708
"	2758	1024 x 8 EPROM	Use as 2708 but pin 21 to +5V, 19 to 0V
TEXAS	4700	1024 x 8 ROM	Use as 2708 but pin 18 must be taken to 0V or +5V depending on chip.
"	2708	1024 x 8 EPROM	Use as Intel 2708
"	2716	2048 x 8 EPROM	Use as Intel 2716 except pin 21 to -5V, pin 19 to +12V, disconnect pin 20 from S1(2) & connect to A10, connect pin 18 to old pin 20 connection to S1(2).
"	2516	2048 x 8 EPROM	Use as Intel 2716
MOTOROLA	68708	1024 x 8 EPROM	Use as 2708
"	68308	1024 x 8 EPROM	Use as 2708, -5V, +12V not needed
"	6830	(e.g. MIKBUG)	Not useable